

Assignment
Deadline: 04 January 2026
Course Title: Artificial Intelligence and Machine Learning Lab
Course Code: STA 0542 5184
Course Instructors:
Dr. Nahid Sultana & Dr. Mohammad Anamul Haque

Instructions:

1. Go to the file “***Data Explanations_AI-ML_Assignment***” and read the dataset descriptions to answer the following questions 1-4.
 2. While preparing your answerscript write and report only relevant informations resulted from your analysis and figures.
 3. Write your registration number at the top of each file and submit the following files in google classroom:
 - I. Output files with relevant tables, graphs, and interpretations in PDF format
 - II. Python script
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Consider ***Credit*** data for **Questions 1-3:**

Question 01:

- a) Load the *Credit* data named “*credit*” into Python using pandas.
- b) Print the dataset.
- c) Create “Education” as feature variable (X) and “Rating” as dependent variable (Y).
- d) Check whether the relationship between X and Y is linear or not.
- e) Split the dataset into training and testing sets, where the training set contains 80% observations.
- f) Fit a simple linear regression (SLR) model of Y on X.
- g) Print OLS regression results, including intercept, regression coefficient, p-value, R^2 , adjusted R^2 and interpret the results.
- h) Use the SLR model to make predictions on the test data and train data, and print the results.
- i) Print model evaluation criteria such as MSE, RMSE, MAE, and interpret.
- j) Visualize the training set results and test set results with a regression line.
- k) Print Q-Q plot of the residuals. How you will interpret this?

Question 02:

Create a matrix X for the feature variables “*Rating, Student, Age, Married, Limit*” and a vector Y for the outcome variable “*Own*”.

- a) Scale the numeric features using standardization technique and create a new feature matrix named X_scaled by keeping the other features unchanged.
- b) Split the dataset of matrix X_scaled and vector Y into training and testing sets, where the training set contains 80% observations.
- c) Classify “Own” by fitting logistic regression, decision tree, random forest, KNN, and SVM classifiers.
- d) Print accuracy, precision, recall, and F1-scores for each classifier and interpret results.
- e) Print the confusion matrix and ROC curve for each classifier, and interpret.

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- f) Print error metrics such as MAE, RMSE for each classifier and interpret.
- g) Which model provides the best “Own” classification in terms of accuracy score?

Question 03:

Create a matrix X for the feature variables “Income, Limit, Age, Own, Student, Married” and a vector Y for the outcome variable “Rating”.

- a) Scale the numeric features using normalization technique, and create a new feature matrix named X_scaled by keeping the other features unchanged.
- b) Split the dataset of matrix X_scaled and vector Y into training and testing sets, where the training set contains 80% observations.
- c) Predict Ratings by fitting decision tree, random forest, KNN, SVM, GradientBoost, AdaBoost, and XGBoost models.
- d) Print the decision tree and interpret.
- e) Print model evaluation metrics such as MSE, MAE, RMSE, and R^2 for each model and interpret.
- f) Which model provides the better prediction in terms of evaluation metrics?
- g) Fit the lasso and ridge regression model.
- h) Print optimal alpha value, MSE, and R^2 for each model and interpret.
- i) Print and identify the best (non-zero) features based on the optimal alpha value for both models.

Question 04:

Consider the *Iris* dataset

- a) Load the *Iris* data into Python using pandas.
- b) Print the dataset.
- c) Perform clustering and dimensionality reduction as follows:
 - i. Apply K-Means clustering and visualize the resulting clusters.
 - ii. Apply a Hierarchical clustering model and visualize the clusters (using an appropriate linkage method).
 - iii. Apply DBSCAN and visualize the identified clusters, including noise points if any.
 - iv. Apply dimensionality reduction techniques such as t-SNE and UMAP and visualize the data in two dimensions, highlighting the class or cluster structure.